

Suppose you are working in a data-driven job role, and you have to answer all these questions.

- First, write a code where, by changing just the keyword, we can search for multiple things.
- Top 15 countries where the keywords are searched the most, and also create visual representation of
- A World map needs to be plotted showing the countries that search the keyword the most.
- We need to extract the time-wise interest of the keyword - how it trended in different years.
- Compare related keywords and plot the graph.

```
In [2]: !pip install pytrends

#It's an unofficial API wrapper for Google Trends
#it tells you what people are searching on Google and how it changes over time

Requirement already satisfied: pytrends in c:\users\sahil\anaconda3\lib\site-packages (4.9.2)
Requirement already satisfied: requests>=2.0 in c:\users\sahil\anaconda3\lib\site-packages (from pytrends) (2.32.5)
Requirement already satisfied: pandas>=0.25 in c:\users\sahil\anaconda3\lib\site-packages (from pytrends) (2.3.3)
Requirement already satisfied: lxml in c:\users\sahil\anaconda3\lib\site-packages (from pytrends) (5.3.0)
Requirement already satisfied: numpy>=1.26.0 in c:\users\sahil\anaconda3\lib\site-packages (from pandas>=0.25->pytrends) (2.4.2)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\sahil\anaconda3\lib\site-packages (from pandas>=0.25->pytrends) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\sahil\anaconda3\lib\site-packages (from pandas>=0.25->pytrends) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\sahil\anaconda3\lib\site-packages (from pandas>=0.25->pytrends) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\sahil\anaconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas>=0.25->pytrends) (1.17.0)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\sahil\anaconda3\lib\site-packages (from requests=2.0->pytrends) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in c:\users\sahil\anaconda3\lib\site-packages (from requests>=2.0->pytrends) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\sahil\anaconda3\lib\site-packages (from requests>=2.0->pytrends) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\sahil\anaconda3\lib\site-packages (from requests>=2.0->pytrends) (2026.1.4)
```

```
In [34]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from pytrends.request import TrendReq
```

Setup pytrend library and keywords define

```
In [92]: pytrends = TrendReq(hl='en-US', tz=360)
Keyword = "Machine Learning"
```

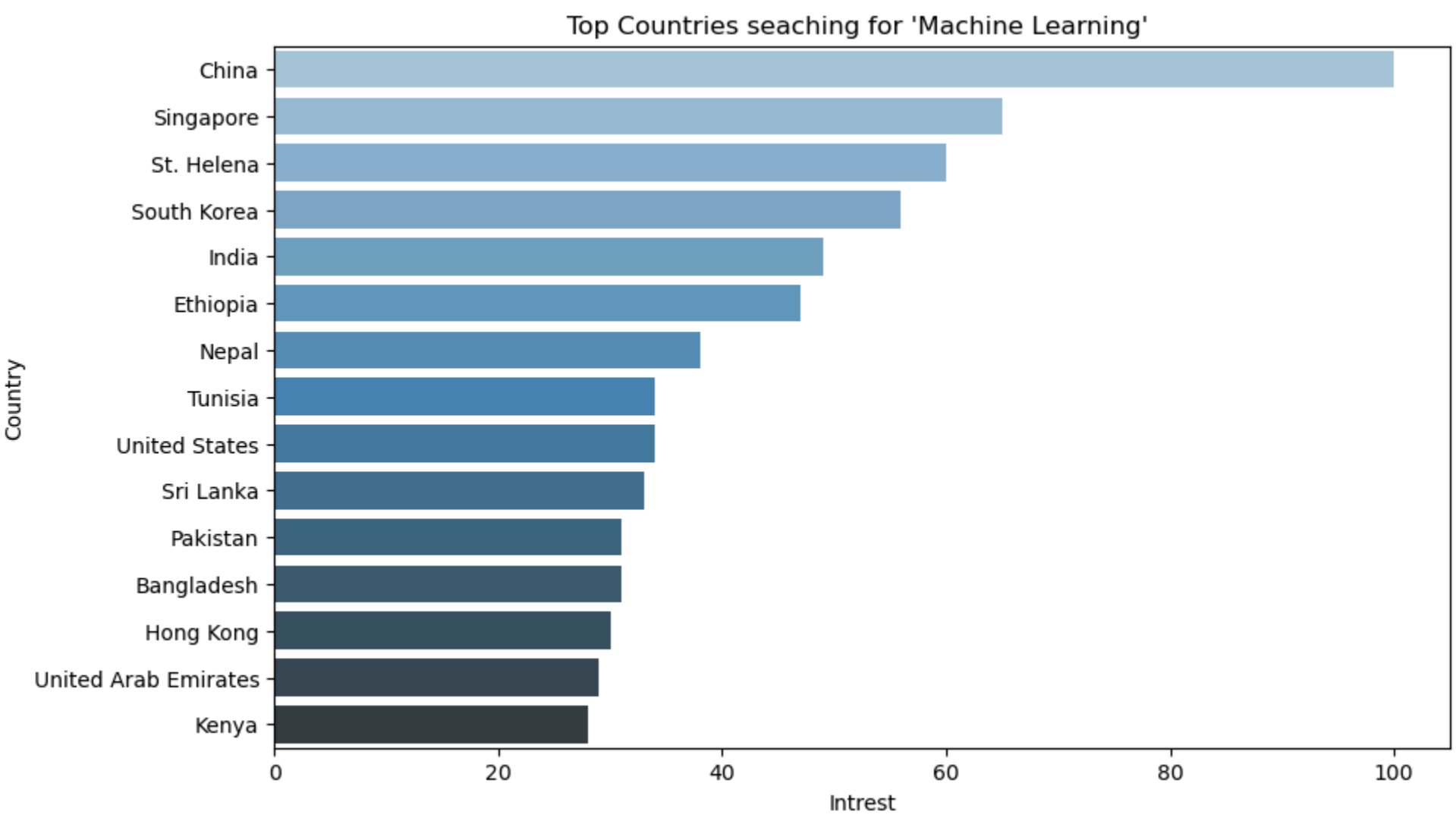
Data Request

```
In [94]: pytrends.build_payload([Keyword], timeframe='today 12-m')
```

Country wise interest

```
In [127]: region_data = pytrends.interest_by_region()
region_data = region_data.sort_values(by=Keyword, ascending=False).head(15)
```

```
In [128]: plt.figure(figsize=(10,6))
sns.barplot(
    x=region_data[Keyword],
    y=region_data.index,
    hue=region_data.index,
    palette="Blues_d",
    legend=False
)
plt.title(f"Top Countries searching for '{Keyword}' ")
plt.xlabel("Intrest")
plt.ylabel("Country")
plt.show()
```



World Map

```
In [132]: region_data = pytrends.interest_by_region()
region_data = region_data.reset_index()
fig = px.choropleth(region_data,
    locations='geoName',
    locationmode='country names',
    color=Keyword,
    title=f"Search Intrest for '{Keyword}' by Country",
    color_continuous_scale='Blues')
fig.show()
```

C:\Users\sahil\AppData\Local\Temp\ipykernel_4188\937838084.py:3: DeprecationWarning:

The library used by the 'country names' 'locationmode' option is changing in an upcoming version. Country names in existing plots may not work in the new version. To ensure consistent behavior, consider setting 'locationmode' to '*ISO-3*'.

Search Intrest for 'Machine Learning' by Country



Time wise Interest

```
In [133]: time_df = pytrends.interest_over_time()
```

C:\Users\sahil\anaconda3\Lib\site-packages\pytrends\request.py:260: FutureWarning:

Downcasting object dtype arrays on .fillna, .ffill, .bfill is deprecated and will change in a future version. Call result.infer_objects(copy=False) instead. To opt-in to the future behavior, set 'pd.set_option('future.no_silent_downcastin g', True)'

```
In [134]: plt.figure(figsize=(12,6))
plt.plot(time_df.index, time_df[Keyword], markers='o', color = 'purple')
plt.title(f"search interest over time '{Keyword}' ")
plt.xlabel('Date')
plt.ylabel('Interest')
plt.grid(True)
plt.show()
```

